Epic 5: Application Building

Description

This phase focuses on transforming the trained machine learning model into a complete, user-friendly web application capable of providing real-time credit card approval predictions. The frontend interface, backend processing logic, trained classification model, and prediction workflow are integrated into a single system. The application allows users to enter applicant information, validates the input data, performs the necessary preprocessing, loads the trained model, generates predictions, and displays the final approval decision through an intuitive web interface.

The application is developed using the Flask web framework. HTML templates, CSS, and Bootstrap are used to create responsive pages that provide a simple and interactive user experience. Flask manages URL routing, handles HTTP requests, processes form submissions, and communicates with the trained machine learning model. The trained model is stored using Joblib/Pickle and is loaded only once during application startup, ensuring fast and efficient predictions without retraining.

Major Components

• Building responsive HTML pages for user interaction.

• Developing the Flask backend to manage routing and request handling.

• Loading the trained machine learning model using Joblib/Pickle.

• Collecting applicant details from web forms and validating inputs.

• Applying the same preprocessing steps used during model training.

• Generating approval or rejection predictions using the trained model.

• Displaying prediction results with a clean and user-friendly interface.

• Testing the complete application locally to verify correct functionality.

• Ensuring smooth communication between the frontend, backend, and machine learning model.

• Preparing the application for future deployment on cloud platforms such as Render, Railway, or Heroku.

Workflow

The user accesses the home page and navigates to the prediction form. After entering applicant details, the data is submitted to the Flask server. The backend validates and preprocesses the input, loads the trained model, performs prediction, and returns the result to the result page. The complete workflow provides fast, accurate, and reliable credit card approval predictions in real time.

Outcome

At the end of this phase, a fully functional end-to-end Credit Card Approval Prediction system is successfully developed. The application integrates machine learning with web technologies, enabling users to obtain instant approval predictions through a simple browser-based interface. This demonstrates successful deployment of the complete machine learning pipeline from data preprocessing and model training to real-time prediction.